Ye Yuan (Mars)

470-884-1922 surprisinglykill@gmail.com www.linkedin.com/in/ye-yuan-468ba8285

Education

Georgia Institute of Technology Expected May 2028
Master of Science in Robotics, College of Computing *Atlanta, Georgia*
- **Relevant Coursework:** Deep Learning for Robotics, Computer Vision, Human Robot Interaction

The Hong Kong Polytechnic University Aug 2022 - May 2026
Internet-of-things with a Secondary Major in Artificial Intelligence and Data Analytics (GPA: 3.96/4.3) *Hong Kong*
- **Relevant Coursework:** Perceptual Robotics, Computer Vision and Pattern Recognition, Optimization Methods, Mobile Edge Computing, Data Structures and Algorithms

Experience

Hand-Held Magnetic Field Generator Alignment System May 2025 – Aug 2025
Mitacs Globalink Research Intern, Microrobotics Lab *Toronto, Canada*
- Designed a perception pipeline using a magnetic sensor array and optimization to locate and align a tool in endoscopic submucosal dissection.
- Implemented Trust Region framework with cosine distance loss function attaining sub-5mm positional and sub-10-degree rotational errors.
- Created a real-time 3D visualization interface to augment surgical capabilities, providing automated rotation axis alignment guidance.

Publications

Adaptive Graph Structure Learning with Node Transformation in Spatial-Temporal Graph Networks for Eating Intention Prediction. *IEEE CASE*, 2026 — Xuan Zhang, **Ye Yuan**, Hailong Huang

Unveiling the Bias Mosaic: Enhancing Doherty Power Amplifiers With Bias Network Variation and Layout Optimization. *IEEE Microwave Magazine*, 2025 — Yanze Wang, Jingyun Bi, **Ye Yuan**, Xinyu Zhou, Wing Shing Chan

Projects

Intention-aware Assistive Feeding Robot | *Python*, *Linux*, *ROS*, *MoveIt*, *Machine Learning*, *Computer Vision*
- Curated and annotated a custom video dataset of over 1000 clips to train and validate intention and emotion recognition models.
- Developed an SMACH state machine to generate intention-aware responses and integrated it with an enhanced STGCN model.
- Implemented eye-to-hand visual servoing with an Intel RealSense camera, ROS, and MoveIt to track the user's mouth and guide feeding motions.

Action Chunking with Transformers Training & Deployment | *LeRobot*, *Git*, *Teleoperation*, *Policy Training*, *3D Printing*
- Assembled SO101 arms and set up teleoperation environment to collect over 60 demonstration episodes.
- Trained an Action Chunking with Transformers (ACT) Policy and achieved 80% success rate in a real-world pick-and-place task.

Vision-based automatic drone detection and tracking | *PyTorch*, *OpenCV*, *YOLO*, *Object Tracking*, *Camera Calibration*
- Improved YOLO small object detection accuracy (COCO AP for small objects) by over 10% by adding a small object detection head.
- Curated a dataset of 15,000 images for model training and enhanced model precision by 3% using hard negatives.
- Utilized ByteTrack with YOLO to achieve 88% detection rate and only 14 ID switches when tracking a drone flying in the forest on a video dataset.
- Built a 3D drone localization system using stereo triangulation and hardware-triggered cameras.

Differential-Drive Mobile Robot Navigation | *ROS2*, *LiDAR*, *Obstacle Avoidance*, *Feedback Control* | *Course Project*
- Integrated camera-based target direction with LiDAR range measurements and implemented separate heading and distance controllers to follow a moving object at a desired distance.
- Developing odometry-based waypoint navigation with LiDAR obstacle detection and reactive avoidance for obstacles encountered along the route.

Technical Skills

Programming & Tools: Python, C, C++, ROS/ROS2, CUDA, Isaac Sim, Mujoco, MATLAB, MongoDB, Conda, Git
Hardware & Embedded Systems: Jetson Nano, Arduino, STM32, ESP32
CAD & 3D Printing: SolidWorks, AutoCAD, Fusion 360, Bambu Studio, Creality Print